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<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Graph Algorithms (course)

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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1
Introduction
and
Principles of
Algorithms ()

Week 2
Shortest
Path
Algorithms
and
problems ()

Week 3
Bellman
Equations ()

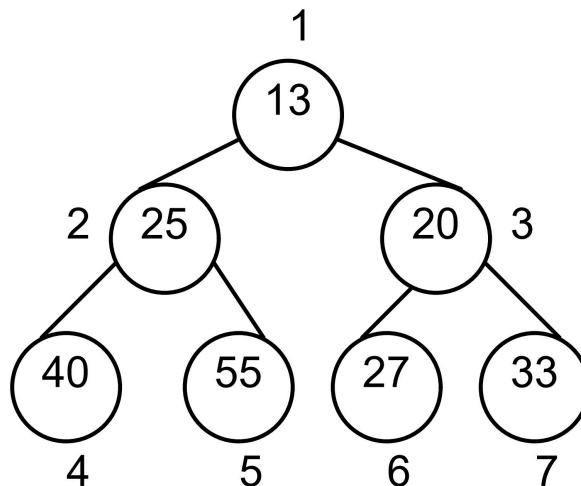
Week 4 Assignment 4

The due date for submitting this assignment has passed.

Due on 2025-08-20, 23:59 IST.

Assignment submitted on 2025-08-20, 19:09 IST

1) Perform Insert (3) and Insert (30) in the given heap. The index of the node containing 40 is ____.



No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: String) 8

1 point

2) Perform Insert (3) and Insert (30) in the given heap. The value at the node with index 9 is ____.

Week 4 Bellman Ford and Dijkstra Algorithm ()

☐ Bellman Ford
P1 (unit?
unit=21&lesso
n=55)

☐ Bellman Ford
P2 (unit?
unit=21&lesso
n=56)

☐ Dijkstra
Algorithm
(unit?
unit=21&lesso
n=57)

☐ Dijkstra
Algorithm P2
(unit?
unit=21&lesso
n=58)

☐ Dijkstra
Algorithm P3
(unit?
unit=21&lesso
n=59)

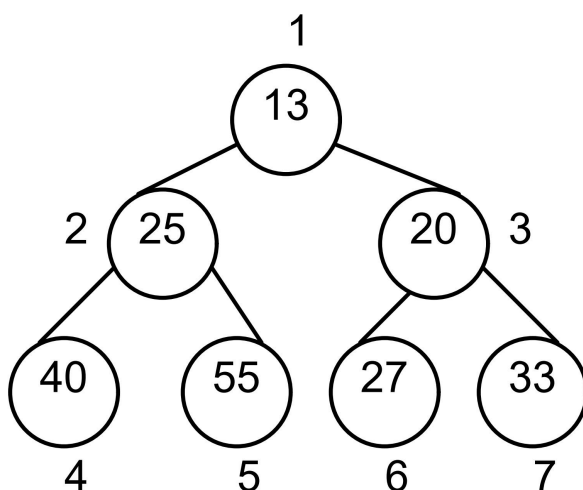
☐ Week 4
Lecture
Material (unit?
unit=21&lesso
n=65)

☒ **Quiz: Week 4
Assignment 4
(assessment?
name=100)**

☐ Feedback
Form (unit?
unit=21&lesso
n=29)

Week 5 All Pair Shortest Path ()

Week 6 Prims and



Yes, the answer is correct.

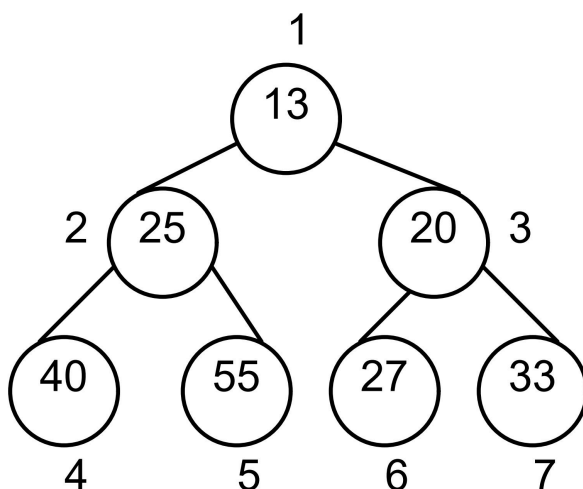
Score: 1

Accepted Answers:

(Type: String) 30

1 point

3) Perform Insert (3) and Insert (30) in the given heap. The value at the root is ____.



Yes, the answer is correct.

Score: 0.5

Accepted Answers:

(Type: String) 3

0.5 points

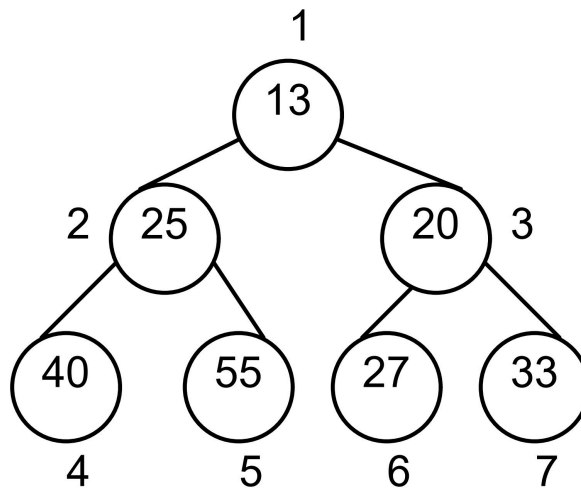
4) Perform Insert (3) and Insert (30) in the given heap. The minimum value in the subtree rooted at the node with index 4 is ____.

Kruskal's
Algorithm ()

Week 7 DFS
and Cut
Vertex ()

Video
download ()

Live session
()



No, the answer is incorrect.

Score: 0

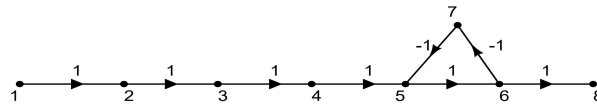
Accepted Answers:

(Type: String) 25

0.5 points

- 5) Execute the Bellman-Ford Algorithm on the following graph. Assume that the vertex 1 is the source

2 points



Consider the edges in the order ((1,2),(2,3),(3,4),(4,5),(5,6),(6,7),(7,5),(6,8))

Array values of $d_0[__]$ are

- ☒ [0, ∞ , ∞ , ∞ , ∞ , ∞ , ∞ , ∞]
- ☐ [∞ , ∞ , ∞ , ∞ , ∞ , ∞ , ∞ , ∞]
- ☐ [∞ , 0, ∞ , ∞ , ∞ , ∞ , ∞ , ∞]
- ☐ [0, ∞ , 0, ∞ , ∞ , ∞ , ∞ , ∞]

Yes, the answer is correct.

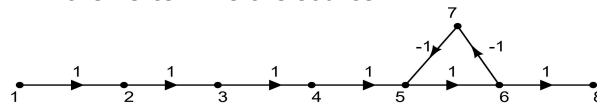
Score: 2

Accepted Answers:

[0, ∞ , ∞ , ∞ , ∞ , ∞ , ∞ , ∞]

- 6) Execute the Bellman-Ford Algorithm on the following graph. Assume that the vertex 1 is the source

2 points



Consider the edges in the order ((1,2),(2,3),(3,4),(4,5),(5,6),(6,7),(7,5),(6,8))

Array values of $d_3[__]$ are

- ☒ [0, 1, 2, 3, ∞ , ∞ , ∞ , ∞]
- ☐ [0, 1, 2, ∞ , 3, ∞ , ∞ , ∞]
- ☐ [0, ∞ , 1, 2, 3, ∞ , ∞ , ∞]

☒ [0,1, ∞, 2,3, ∞, ∞, ∞]

Yes, the answer is correct.

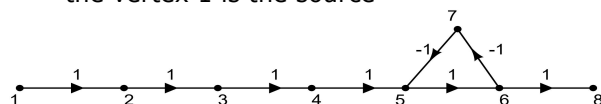
Score: 2

Accepted Answers:

[0,1, 2,3, ∞, ∞, ∞, ∞]

- 7) Execute the Bellman-Ford Algorithm on the following graph. Assume that the vertex 1 is the source

2 po



Consider the edges in the order ((1,2),(2,3),(3,4),(4,5),(5,6),(6,7),(7,5),(6,8))

Array values of d_7 [] are

☒ [0,1, 2,3, 4,5, 4,6]

☐ [0,1, 2,3, 2,5, 4,6]

☐ [0,1, 2,2, 3,5, 4,6]

☐ [0,1, 2,3, 3,5, 4,6]

No, the answer is incorrect.

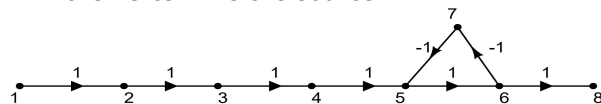
Score: 0

Accepted Answers:

[0,1, 2,3, 3,5, 4,6]

- 8) Execute the Bellman-Ford Algorithm on the following graph. Assume that the vertex 1 is the source

2 points



Consider the edges in the order ((1,2),(2,3),(3,4),(4,5),(5,6),(6,7),(7,5),(6,8))

Array values of d_8 [] are

☐ [0,1,2,3,3,4,4,6]

☐ [0,1,2,2,3,4,4,6]

☒ [0,1,2,3,4,4,4,6]

☐ [0,1,2,2,3,4,4,6]

No, the answer is incorrect.

Score: 0

Accepted Answers:

[0,1,2,3,3,4,4,6]